



CS 508: Introduction to Heterogeneous Computing

Course Code	CS 508
Course Name	Introduction to Heterogeneous Computing
L-T-P-C	2-0-0-2
Prerequisites	Introductory Programming skill
Intended for	MS/MTech/PhD/I-PhD/4th year BTech
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **Review of programming concepts:** C/C++: Algorithm design, Expressions, Decisive statements, Iteration loops, Functional calls, Recursion, Arrays, Pointers, Addresses, I/O file handling, Error handling [6 Lectures]
- **Introduction to HPC:** Introduction to uni-processor and multiprocessor architecture, types of parallelism, Data parallelism, Loop unrolling, communication and synchronization needs, Parallel programs using openMP/ MPI, Introduction to cluster hardware, software, and network [8 Lectures]
- **Understanding parallelism with GPUs:** CUDA C: GPU architectures, Grids, Blocks and Threads in CUDA, Memory Handling with CUDA, Concepts of tiling and warps for CUDA [12 Lectures]
- **Application case studies:** Problem statements, designing an algorithm, writing program, verifying the output profiling the performance, optimization of the program [8 Lectures]

Text Books

1. Cook S., **CUDA Programming**,
2. Morgan Kaufmann, **nVIDIA GPU Teaching Kit**.
3. Kirk D.B. and Hwu W-M.W., **Programming Massively Parallel Processors**.